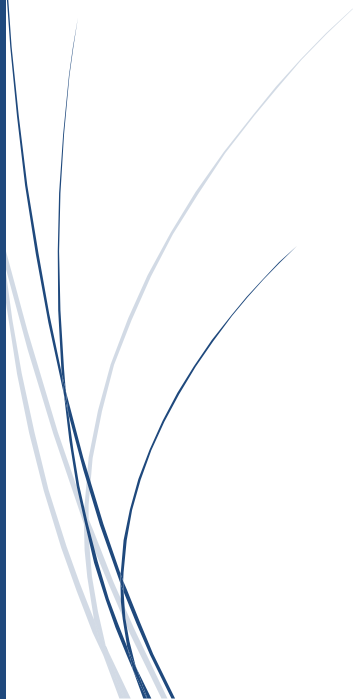


Test Project

Module 4

Mobile Applications Development

Day 2 P.M



Contents

Contents 2

Introduction..... 3

Description of project and tasks..... 3

Instructions to the Competitor13

Introduction

In the ongoing pursuit of fostering cognitive development through engaging experiences, Zoology is proud to unveil "Memory Game" a groundbreaking mobile application. Memory game" reimagines the classic memory match game, infusing it with the captivating world of animals. This innovative approach delivers a fun and educational experience for users of all ages, fostering memory skills while igniting a love for the natural world.

The game primary involves players flipping open covered cards and trying to match identical animal cards as fast as possible. At the end of the game, the application will display the user's a historical ranking leaderboard. The user is allowed to end the game prematurely or restart the game by doing a phone shake or by navigating through the application.

The product manager has provided you with product wireframes. As a developer of mobile applications, your task is to develop corresponding apps based on the given wireframes according to specific development needs.

Description of project and tasks

The developer does not need to consider the aesthetics of the program interface and the position of the elements. The work can be inconsistent with the given wireframes, but the developer needs to properly consider the ease of use. The development task is divided into two parts depending on the devices.

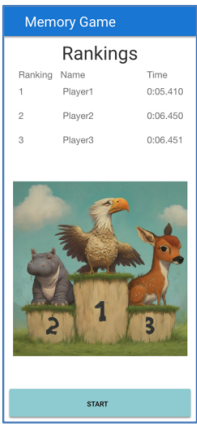
Part	Name	Competition Time	Marking Devices
A.M.	Functionality (phone)	2.5h	Emulator: Pixel 2 or iPhone 8 Plus

Application wireframes

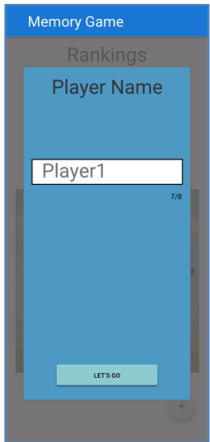
The App wireframes for reference only. You do not need to copy the same appearance, colors, decorative icons/images, or elements' positions/style.



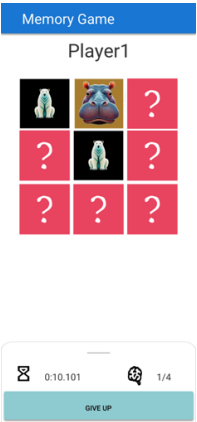
Splash screen



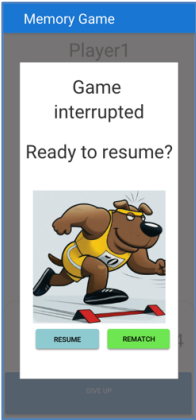
Rankings



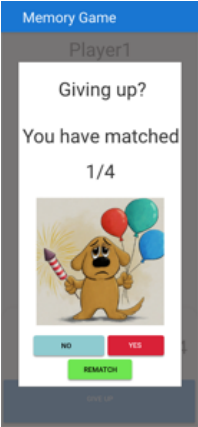
Player Name



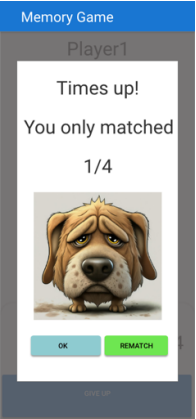
Game play



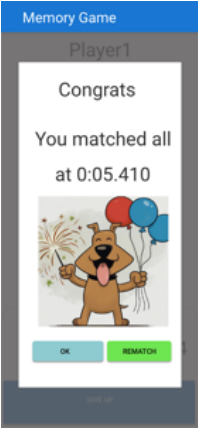
Game pause – Ready to Resume



Game pause - Give up



Game end - Time Expired



Game end - Completed

General Demands

1. Finish the function of this application according to the demands.
2. You need to make this application easy to use.

Pages Demands

1 Splash Screen page

1.1 Elements include:

- Title "Memory Game"
- Background image

1.2 Functional requirements:

- Splash screen will appear for 2 seconds.
- Navigates to "Rankings" page after Splash screen expires.
- Animated appearance of title.



Pages Demands

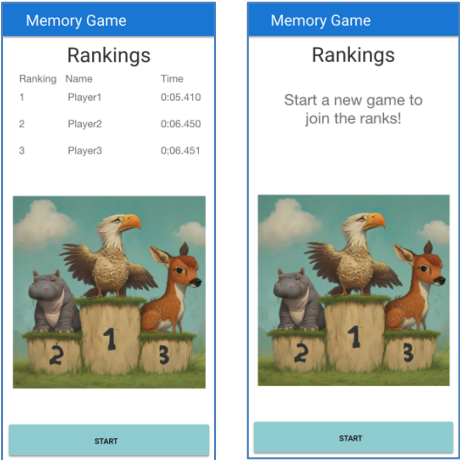
2 Rankings page

2.1 Elements include:

- Title "Rankings"
- 3 columns
 - "Ranking"
 - Displays "1", "2", "3"
 - "Name"
 - Display user's name
 - "Time"
 - Format: mm:ss.sss
- "Start a new game to join the ranks!" text.
- Static image.
- Start button
 - Color: #8ECBD0

2.2 Functional requirements:

- Displays the fastest 3 that played the game on this device.
- Ranking is sorted in ascending order.
- New entry will have an animated display.
- "Start a new game to join the ranks!" to display when there is no past completed game.
- "Start" will navigate user to "Player Name" page.
- Scores will persist until the application is either uninstalled or user performs data purge.



Pages Demands

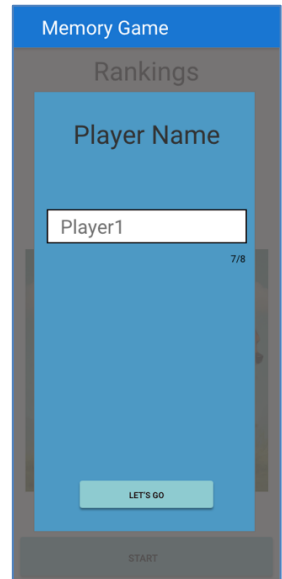
3 Player Name page

3.1 Elements include:

- Title "Player Name"
- Text box for user to key in name
- Character count
- "LET'S GO" button
 - Color: #8ECBD0

3.2 Functional requirements:

- Display "Enter user name here" when the textbox is empty.
- Allows user to key in name, with a maximum of 8 characters.
- Character count decrease and increase according to user input.
- Rankings screen to be partially covered and shadowed over.
- Taping on "LET'S GO" button will navigate user to the "Game Start" page.
- Background color: #4D99C4



Pages Demands

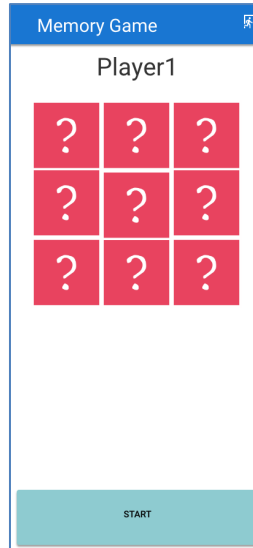
4 Game Start page

4.1 Elements include:

- Player's name.
- 3 x 3 set of "?"
- Start button
 - Color: #8ECBD0
- Exit game option

4.2 Functional requirements:

- "?" cards will remain until user taps on "START" button.
- Tapping Exit game will navigate user back to "Rankings" page.
- Tapping Start button will start the game.

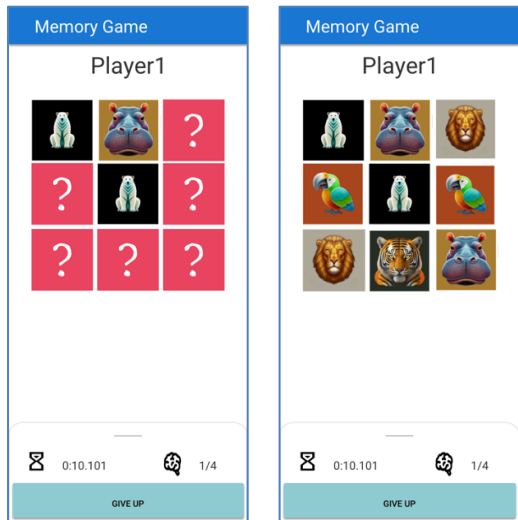


Pages Demands

5 Game Play page

5.1 Elements include:

- Player's name
- 3 x 3 cards
- Game panel
 - Timer icon
 - Pairs found icon
 - Timer
 - Format mm:ss.sss
 - "GIVE UP" button
 - Color: #8ECBD0



5.2 Functional requirements:

- Game panel will slide up when user navigates into this page.
- Timer starts and updates the timer text until the game expires.
- 9 cards should consist of 4 pairs of cards and 1 random extra card.
- Positions of cards should be random and different from previous game.
- Only two cards can be selected at each attempt.
 - Matched pair will remain open.
 - Unmatched pair will close the cards opened during the attempt.
- Game expires when it exceeds the last timing in leader board. User will be navigated to "Time is up" page.
- If no time exists, the game will not expire.
- Tapping the "GIVE UP" button will navigate the user to the "Give up screen".
- Shaking the device will trigger the "GIVE UP" button.
- When the user navigates away during the game, the game will be paused and display "Ready to resume" page when the user navigates back to the application.
- Successful matching all pairs before the time expires will navigate user to "Completed" page.

Pages Demands

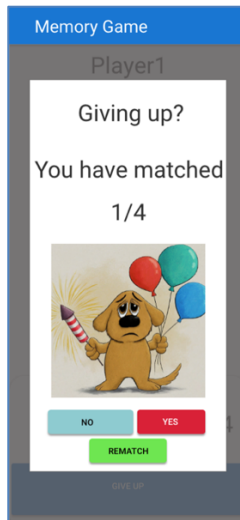
6 Game Pause page

6.1 Elements include:

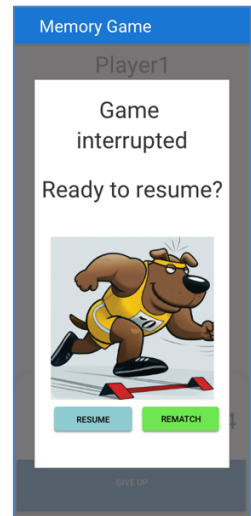
- Title text.
 - “Giving up?”
 - “Game interrupted”
- Message text
 - “You have matched”
 - “Ready to resume”
- Number of pairs matched text [Give up page ONLY].
- Static image.
- “YES” button [Give up page ONLY]
 - Color: #D92137
- “NO” button [Give up page ONLY]
 - Color: #8ECBD0
- “RESUME” button [Ready to resume page ONLY]
 - Color: #8ECBD0
- “REMATCH” button [Give up and Ready to resume page]”
 - Color: #6EE651

6.2 Functional requirements:

- Game pause screens will display to the user in two occasions:
 - Toggle of “Give up” button during game play.
 - User is navigated away from the application by other applications, e.g. phone call, during game play.
- In the “Give up” page,
 - Number of pairs matched text will display how many pairs the user has found out of the 4 pairs given.
 - e.g. 1/4, 2/4, 3/4
- Tapping on “YES” will navigate user to “Rankings” page.
- Taping on “REMATCH” will navigate user to the start of a new game.
- Tapping on “NO” will resume game.
- Game screen to be partially covered and shadowed over.



Give up page



Ready to resume page

Pages Demands

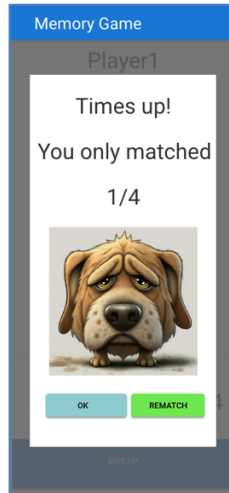
7 Game End page

7.1 Elements include:

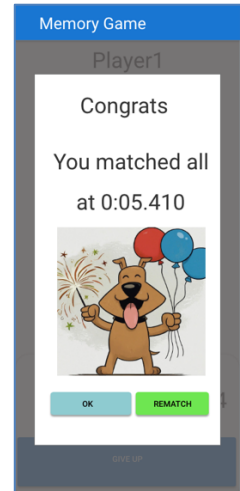
- Title text.
 - "Times up!"
 - "Congrats"
- Message text
 - "You only matched"
 - "You matched all"
- Number of pairs matched text ["Only for Time is up" page].
- Time taken text [Only for "Completed" page].
- Static image.
- "OK" button.
 - Color: #8ECBD0
- "REMATCH" button
 - Color: #6EE651

7.2 Functional requirements:

- Game end screens will display to the user in two occasions:
 - Expiration of time.
 - User completed matching.
- In the "Time is up" page,
 - Number of pairs matched text will display how many pairs the user has found out of the 4 pairs given.
 - e.g. 1/4, 2/4, 3/4
- Tapping on "OK" will navigate user to Leaderboard.
- Taping on "REMATCH" will navigate user to the start of a new game.
- Screen expires in 2 seconds and navigates user to Leaderboard.
- Game screen to be partially covered and shadowed over.



Time is up page



Completed page

Instructions to the Competitor

- 1) Please create the project with correct package name (Android)/ Organization Identifier (iOS). Naming convention: edu.ws2024.aXX.pm;
- 2) You should save the project in the folder: XX_Module_A_PM;
- 3) You should rename the generated apk (Android)/ app (iOS) file as XX_Module_A_PM.apk (Android)/ XX_Module_A_PM.app (iOS), and save it in the root of the XX_Module_A_PM folder;
- 4) The whole XX_Module_A_PM folder should be pushed to the provided remote Git repository;
- 5) Submit part A.M. in the morning and part **P.M.** in the afternoon, and the two parts will be marked independently;
- 6) Note: XX refers to your workstation code.